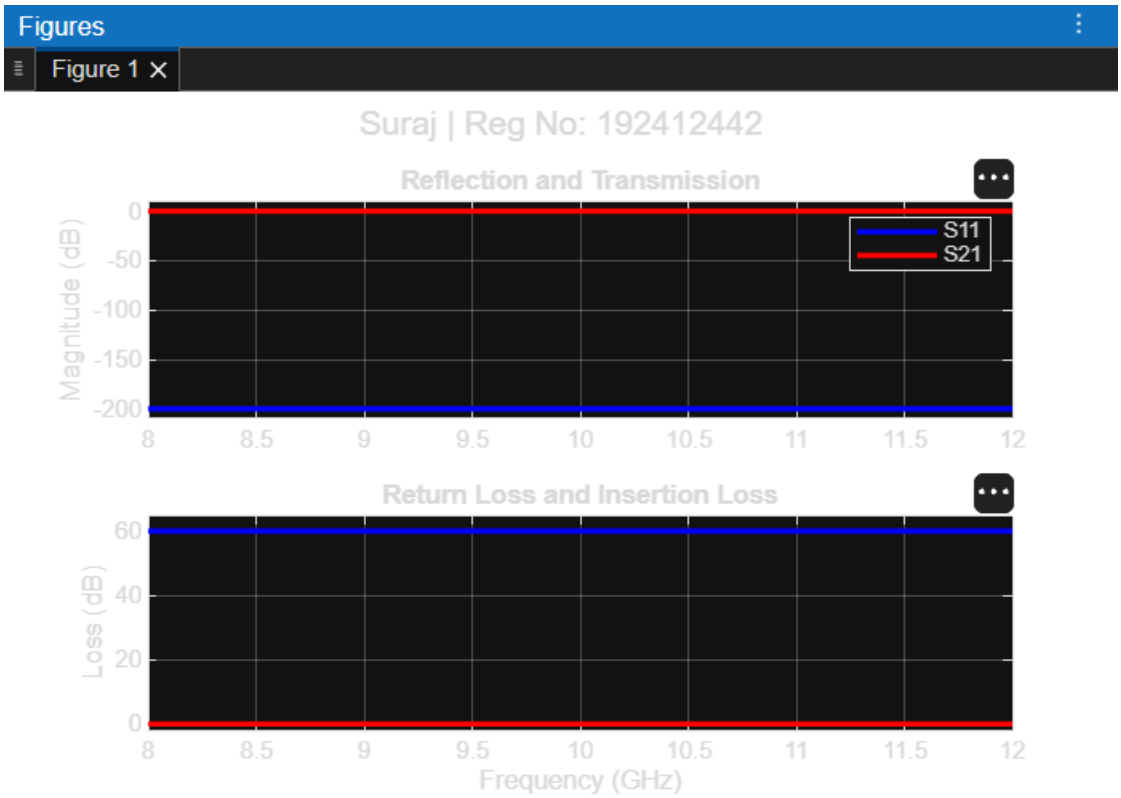
EXPERIMENT 2 (SURAJ NARASIMHAN POTTY_192412442)

CODE:

```
clc; clear; close all;
c=3e8; Z0=50; ZL=50; L=0.05;
F=linspace(8e9,12e9,201);
beta=2*pi*F/c;
G=(ZL-Z0)/(ZL+Z0);
S11=G*ones(size(F)); S21=exp(-1j*beta*L);
RL=-20*log10(max(abs(S11),1e-10));
IL=-20*log10(abs(S21));
fprintf('Suraj | Reg No: 192412442\n');
fprintf('Wavelength = %.1f mm\n',c/10e9*1e3);
fprintf('Beta = %.2f rad/m\n',2*pi*10e9/c);
fprintf('S11 = %.2f | S21 = %.2f\n',abs(G),abs(S21(101)));
fprintf('Return Loss = Inf dB | Insertion Loss = 0 dB\n');
figure('Color','w','Position',[100 100 900 600]);
subplot(2,1,1); plot(F/1e9,20*log10(max(abs(S11),1e-10)),'b','LineWidth',2);
hold on; plot(F/1e9,20*log10(abs(S21)),'r','LineWidth',2);
grid on; ylim([-210 10]); ylabel('Magnitude (dB)');
legend('S11','S21'); title('Reflection and Transmission');
subplot(2,1,2); plot(F/1e9,min(RL,60),'b','LineWidth',2);
hold on; plot(F/1e9,IL,'r','LineWidth',2); grid on;
ylim([-2 65]); xlabel('Frequency (GHz)'); ylabel('Loss (dB)');
title('Return Loss and Insertion Loss');
sgtitle('Suraj | Reg No: 192412442');
```

OUTPUT:


Figures
Figure 1
Suraj | Reg No: 192412442
Reflection and Transmission
Magnitude (dB)
0
-50
-100
-150
-200
S11
S21
8
8.5
9
9.5
10
10.5
11
11.5
12
Return Loss and Insertion Loss
Loss (dB)
60
40
20
0
Frequency (GHz)